Class Test 2: **CS366**	Duration: **30m** Max-Marks: **30** Date: **19/02/2025**	Roll: Name:

[Q1] How many value assignments to a CNF clause containing n literals (one for each binary variable) make the clause true?

[A] 2^n [B] $2^n - 1$
[C] 2^{n-1} [D] $2^{n-1} - 1$

[Q2] If S is a set of clauses that resolve to a clause R then

[A] $S \models R$ [B] $R \models S$
[C] $R \in S$ [D] None of the above

[Q3] Fill in the blanks: An inference algorithm that derives only entailed sentences is called _______ and if it can derive any sentence that is entailed it is called _________.

[A] complete, sound [B] sound, truth-preserving
[C] sound, complete [D] complete, truth-preserving

[Q4] Fill in the blanks: ____ clause is a disjunction of literals of which exactly one is positive whereas ______ clause is disjunction of literals of which at most one is positive.

[A] Goal, Horn [B] Horn, Definite
[C] Definite, Goal [D] Definite, Horn

[Q5] Fill in the blanks: Backward chaining is a form of ______.

[A] Backtracking [B] Logical reasoning
[C] Data-Driven reasoning [D] Goal-Directed reasoning

[Q6] Which of the following is not a valid (necessarily true) sentence?

[A] (∃x x=x) ⇒ (∀y ∃z y=z) [B] ∀x P(x) ∨ ¬P(x)
[C] ∀x Smart(x) ∨ (x=x) [D] ∃x P(x)

[Q7] Use this truth table to select the correct statement:

p	q	r	$p \Rightarrow q \vee r$	$p \Rightarrow r$	$q \Rightarrow r$
1	1	1	1	1	1
1	1	0	1	0	0
1	0	1	1	1	1
1	0	0	0	0	1
0	1	1	1	1	1
0	1	0	1	1	0
0	0	1	1	1	1
0	0	0	1	1	1

[A] $\{p \Rightarrow q \vee r\} \models (p \Rightarrow r)$
[B] $\{q \Rightarrow r\} \models (p \Rightarrow q \vee r)$
[C] $\{p \Rightarrow q \vee r, p \Rightarrow r\} \models (q \Rightarrow r)$
[D] $\{p \Rightarrow q \vee r, q \Rightarrow r\} \models (p \Rightarrow r)$

[Q8] Let Γ and Δ be sets of sentences in Propositional Logic and let ϕ and ψ be individual sentences in Propositional Logic. Select the correct statement:

[A] If $\Gamma \models \phi$ and $\Delta \models \phi$, then $\Gamma \cap \Delta \models \phi$ [B] If $\Gamma \models \varphi$ and $\Delta \models \varphi$, then $\Gamma \cup \Delta \models \varphi$
[C] If $\Gamma \not\models \psi$, then $\Gamma \models \neg\psi$ [D] If $\Gamma \models \neg\psi$, then $\Gamma \not\models \psi$

[Q9]. In propositional logic, the resolution rule is used primarily for:

[A] Converting formulas to CNF [B] Proving satisfiability
[C] Simplifying expressions [D] Deductive inference in refutation proofs

[Q10]. Which first-order logic formula correctly expresses “Every student loves logic” using S(x) for “x is a student” and L(x) for “x loves logic”?

[A] $\exists x\,(S(x) \wedge L(x))$ [B] $\forall x\,(S(x) \rightarrow L(x))$

[C] $\forall x\,(L(x) \rightarrow S(x))$ [D] $\exists x\,(L(x) \rightarrow S(x))$

[Q11]. Which of the following best describes the purpose of the unification algorithm in first-order logic?

[A] It determines whether two formulas are logically equivalent.

[B] It finds a substitution that makes two expressions identical.

[C] It converts a formula into CNF.

[D] It eliminates redundant quantifiers from a formula.

[Q12]. Which statement correctly distinguishes between function symbols and predicate symbols in first-order logic?

[A] Function symbols return truth values, while predicate symbols return objects.

[B] Function symbols return objects, while predicate symbols return truth values.

[C] Both function symbols and predicate symbols return objects.

[D] Both function symbols and predicate symbols return truth values.

[Q13]. What is the typical stopping condition used in value iteration?

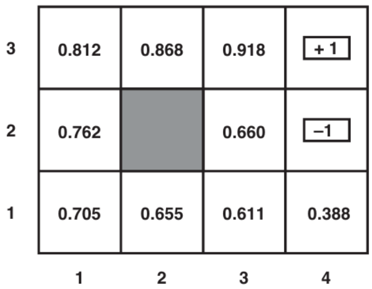
[A] The policy remains unchanged over two successive iterations.

[B] The maximum change in the value function across all states is less than a predetermined threshold.

[C] A fixed number of iterations is reached, regardless of value changes.

[D] All state-values become identical.

[Q14]. Consider the grid-world as shown in the diagram below. What is the probability that an agent starting in state (1,1) reaches state (3,2) by taking an action sequence (*Up, Up, Right*)?



[A] 0.001

[B] 0.008

[C] 0.512

[D] 0.073

[Q15]. In value iteration, what is the primary update performed at each iteration for each state?

[A] Compute the policy’s probability distribution over actions.

[B] Update the state’s value by taking the maximum over actions of the expected return.

[C] Update the state’s value by averaging over all actions.

[D] Update the state’s policy by sampling actions from the current policy.